



Digital storage oscilloscope by Kusam Electrical Industries

One important implication for test equipment going forward is that the expertise to validate channel compliance needs to be embedded into the instruments. Besides, advanced serial protocols are likely to make probing more challenging.

Video. In the video industry, T&M will continually evolve as new technologies arrive and picture/audio specifications and delivery systems change. It will be a whole new ballgame with digital video and its multiple formats, compression methods, screen sizes and other variations. The transition to higher-resolution digital video requires a new class of test and measurement tools for broadcasters to accommodate newer, faster and more complex technologies such as HD, 3G-SDI and IP.

Semiconductor chips. Keeping up with the design trends is pretty challenging for T&M companies. This eventually means designing ever-more capable front-end ICs for next-generation instruments.

The steadily shrinking prices of microprocessors make it financially feasible to incorporate multiple microprocessors into the design of inexpensive instruments. In this way, each different instrument function (display, analogue-to-digital conversion, etc) has its own dedicated microprocessor, allowing far higher throughput as no function has to wait its turn for processing by a central processor.

At the same time, as the worldwide demand of semiconductors for cell-

phones, portable wireless devices and consumer electronics grows, T&M designers have to offer instruments capable of testing the increasingly sophisticated components that go into these devices.

T&M design trends

Increasingly complex testing requirements are driving T&M manufacturers to improve and incorpo-

rate features like multichannel testing, higher bandwidth and smoother PC interface with user-friendly software. Transition from analogue to digital, and from standalone instrument to a total test solution, and development of application-specific test instruments are the primary transitions.

Driven by the user need, there is transition towards hybrid test equipment that occupy less space but perform multiple functions. Test equipment are now no more mere benchtop. These are increasingly becoming integrated with PCs for real-time measurements and monitoring by adopting PXI (peripheral extension interconnect), PCI (peripheral component interconnect) PC-based platform and VXI technologies.

Software-defined instrumentation based on National Instruments'

LabVIEW graphical programming platform, parallel processing technologies, and new methods for wireless and semiconductor tests help engineers develop faster and more flexible automated test systems.

Also, T&M instruments today incorporate highly graphical displays/readouts and are easier to interface with external ancillary equipment such as part handlers, probes and switches. Use of new communication interfaces and smart switching is changing the way they are used.

New communication interfaces. Until recently, instruments employed GPIB and serial (RS-232) interfaces for PC communications. Today, however, most system instruments include Ethernet, or better yet LXI and USB interfaces, in addition to the legacy buses, which may be provided as optional or standard equipment.

"Over time, higher-performing and lower-cost interfaces will win out over legacy interfaces such as GPIB on rack-and-stack instruments. But this displacement is likely to take years to transpire," anticipates Klaus Leutbecher, vice president-worldwide sales, Keithley Instruments.

Smart switching. Manufacturers of switching hardware for automated testing have been talking up this concept for production test applications for close to two decades. Now some new designs employ an embedded test script processor (TSP) that offers enhanced capabilities for controlling test sequencing/flow, decision-making and instrument autonomy.

"Users program and communicate with TSP-enabled instruments in two ways—either by executing individual TSP commands (similar to sending individual SCPI commands) or by writing test scripts. Designed to reside on the instrument or switch mainframe itself, such scripts are a collection (list) of instrument control commands and/or program



MSO 5204 mixed-signal oscilloscope by Tektronix